

```

// RECURSIVE CASE
let powerSet = [];
let head = chars[0];
let tail = chars.slice(1, chars.length);

// First part, get the sets that don't include the head:
document.write(debugMsg +
" part 1, get sets without head '" + head + "'<br />");
let tailPowerSet = getPowerSet(tail, indent + 1); ❷

// Second part, get the sets that include the head:
document.write(debugMsg +
" part 2, get sets with head '" + head + "'<br />");
for (tailSet of tailPowerSet) {
    document.write(debugMsg + " New set " + head + tailSet + "<br />");
    powerSet.push(head + tailSet); ❸
}

powerSet = powerSet.concat(tailPowerSet);
document.write(debugMsg + " returning " + powerSet + "<br />");
return powerSet; ❹
}

document.write("<pre>The power set of ABC:<br />")
document.write(getPowerSet("ABC") + "<br /></pre>");
</script>

```

The programs output the following:

```

The power set of ABC:
In getPowerSet("ABC"), start.
In getPowerSet("ABC") part 1, get sets without head 'A'
.In getPowerSet("BC"), start.
.In getPowerSet("BC") part 1, get sets without head 'B'
..In getPowerSet("C"), start.
..In getPowerSet("C") part 1, get sets without head 'C'
...In getPowerSet(""), start.
...In getPowerSet("") base case returns ['']
..In getPowerSet("C") part 2, get sets with head 'C'
..In getPowerSet("C") New set C
..In getPowerSet("C") returning ['C', '']
.In getPowerSet("BC") part 2, get sets with head 'B'
.In getPowerSet("BC") New set BC
.In getPowerSet("BC") New set B

```